CSE440: Natural Language Processing II

Lab Assignment 2

1. Download the Amazon Customer-Reviews Polarity and take the first 25,000 rows and last 25,000 rows from the train set. Drop the '**title**' column and convert the '**content**' texts to lowercase and remove punctuation. Then, use CountVectorizer to convert the preprocessed texts into Bag-of-Words feature vectors. Report the dimensions of the resulting feature matrix.

2. Using the same preprocessed dataset, apply TfidfVectorizer to generate TF-IDF embeddings. Identify the top 30 words with the highest TF-IDF scores.

3. Obtain the GloVe embeddings (glove.6B.50d.txt). Perform analogy tasks such as "Programmer - Man + Woman" and check whether the resulting vector is closest to "Doctor" using the GloVe embeddings.

4. Select Brown corpus and load the text data. Preprocess the text by tokenizing, converting to lowercase. Train Word2Vec models using both Skipgram and CBOW on your chosen corpus (gensim.models.Word2Vec). Evaluate the trained models on the same analogy task from the previous question (if you encounter any error on the same analogy task, you may come up with any suitable analogy to test).